IS6400 Business Data Analytics

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City University of Hong Kong

Lecture 9

- **Review**
- **Autoregression**

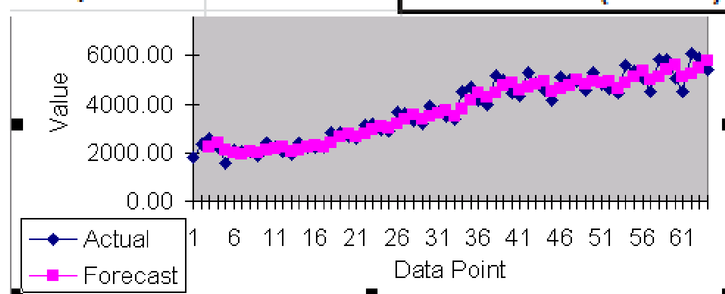
Reading Material:

Simple Moving Average

- *Average random fluctuations in a time series* to infer short-term changes in direction
- **Assumption**: future observations will be similar to recent past
- Forecast = average of most recent k observations

$$F_{t+1} = \frac{Y_t + \ldots + Y_{t-m+1}}{m}$$

Quarter	Sales	F
Q1-94	1807.37	
Q2-94	2355.32	
Q3-94	2591.83	
Q4-94	2236.39	=AVERAGE(B2:B4)



- So, how do you decide m?
  What if m=1 or 2?

Weighted Moving Average

- A natural extension to moving average.
  - Weight the most recent k observations, with weights that add to 1.0
  - *Higher weights on more recent observations generally provide more responsive forecasts to rapidly changing time series*
  - Rational: recent observation are more similar as what will happen next

$$F_{t+1} = \alpha_1 Y_t + \ldots + \alpha_k Y_{t-m+1}$$

$$\alpha_1 + \ldots + \alpha_m = 1$$

Exponential Smoothing

- So: an easier approach (with one parameter)
  - Weights decay exponentially
  - Sum of weights approach 1

$$F_{t+1} = \alpha Y_t + \alpha(1-\alpha)Y_{t-1} + \alpha(1-\alpha)^2 Y_{t-2} + \alpha(1-\alpha)^3 Y_{t-3} + \ldots$$

- Exponential smoothing model:

$$F_{t+1} = L_t = (1-\alpha)L_{t-1} + \alpha Y_t$$
$$= L_{t-1} + \alpha(Y_t - L_{t-1})$$

last prediction + adjustment on error

- F_{t+1} is the forecast for time period t+1,
- L_t is the level for period t, ($L_1=Y_1$)
- Y_t is the observed value in period t, and

α is a constant between 0 and 1, called the smoothing constant.

Higher smoothing constant (higher weights on more recent observations) provide forecasts that respond more rapidly to changes in series.

Holt-Linear

Simple exponential smoothing + trend

- When there is a trend in the series, Holt's method deals with it explicitly by including a trend term, T_t, and a corresponding smoothing constant $\mathcal{B}$.
- L_t : The level of time series.
- T_t : An estimate of the change in the series from one period to the next.
- Formulas for Holt's exponential smoothing method:

$$L_t = \alpha Y_t + (1-\alpha)(L_{t-1} + T_{t-1}) \longrightarrow F_t$$

$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1}$$

$$F_{t+k} = L_t + kT_t$$

Increase in the trend

Holt-Winter

Simple exponential smoothing + trend + seasonality

- Similar to Holt's model, but also has seasonal indexes and a corresponding smoothing constant γ (gamma).
- The smoothing constant gamma controls how quickly the method reacts to observed changes in the seasonality pattern.

$$L_t = \alpha \frac{Y_t}{S_{t-M}} + (1-\alpha)(L_{t-1} + T_{t-1})$$

Trying to estimate Y/S

Seasonal index

$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1}$$

$$S_t = \gamma \frac{Y_t}{L_t} + (1-\gamma)S_{t-M}$$

Trying to estimate S using Y/L

# of season

$$F_{t+k} = (L_t + kT_t)S_{t+k-M}$$

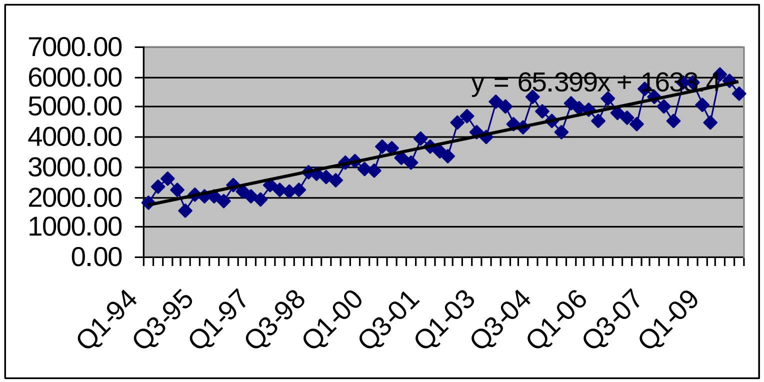
Additive trend times multiplicative seasonal effect

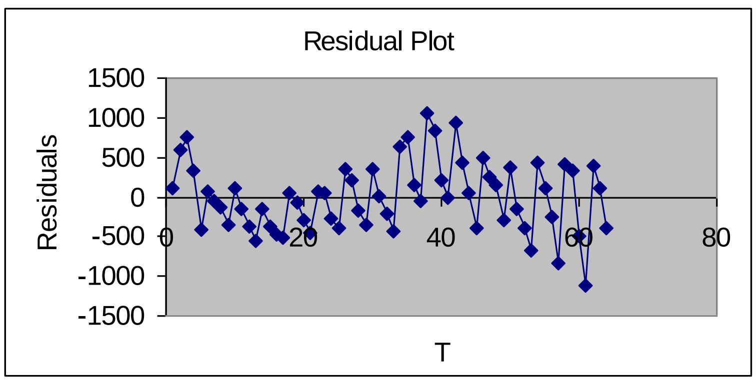
Regression on Linear Trend

- A linear trend means that the time series variable changes by *a constant amount* each time period.
- Linear trend model is given by: $\boldsymbol{Y_t = a + bt + e_t}$
- Interpretation of *b* is that it represents the expected change in the series from one period to the next.
  - If *b* is positive, the trend is upward.
  - If *b* is negative, the trend is downward.
- The intercept term *a* is less important: It literally represents the expected value of the series at time 0.
- As always, a graph of the time series is a good place to start. It may indicate whether a linear trend is likely to provide a good fit.

Soft Drink Sales

- $Sales_t = a + b*t + e_t$.





SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.941304					
R Square	0.886054					
Adjusted R	0.884216					
Standard E	440.1709					
Observatic	64					
ANOVA						
	df	*SS*	*MS*	*F*	*ignificance F*	
Regressior	1	93410374	93410374	482.117	6.12E-31	
Residual	62	12012527	193750.4			
Total	63	1.05E+08				
	Coefficients	*andard Err*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	1633.356	111.3451	14.66931	7.85E-22	1410.78	1855.931
T	65.39902	2.978482	21.95716	6.12E-31	59.44512	71.35292

Regression on Exponential Trend

- An exponential trend is appropriate when the time series changes by a constant percentage (as opposed to a constant amount) each period.
- Appropriate regression equation contains a multiplicative error term u_t: $Y_\mathrm{t} = ce^{bt}u_t$.
- Exponential trend is not useful for estimation, and a linear equation is required. You can achieve linearity by taking natural logarithms of both sides of the equation: $\boldsymbol{ln(Y_t) = a + bt + e_t}$.
- Interpreting final result: *The coefficient b (expressed as a percentage) is approximately the percentage change per period.* For example, if b = 0.05, then the series is increasing by approximately 5% per period.

Regression on Seasonal Effect

- A regression approach to forecasting seasonal data **uses dummy variables for the seasons**.
  - For quarters, create three dummy variables, and the coefficients of the dummy variables, b_1, b_2, and b_3, indicate how much each quarter differs from the reference quarter.
- Depending on how the regression equations are written, you can create either an additive or a multiplicative seasonal model.

Soft Drink Sales

- Consider both trend and seasonal effect
- $\text{LogSales}=b_0+b_1t+b_2Q_1+b_3Q_2+b_4Q_3$
  - The figure illustrates the data setup, with dummy variables for quarters and a transformed *sales* (*Log*(*sales*)) variable.
  - Regression is straightforward.

	A	B	C	D	E	F	G
1	Quarter	Sales	Time	Q1	Q2	Q3	Log(Sales)
2	Q1-94	1807.37	1	1	0	0	7.499628
3	Q2-94	2355.32	2	0	1	0	7.7644319
4	Q3-94	2591.83	3	0	0	1	7.8601195
5	Q4-94	2236.39	4	0	0	0	7.7126182
6	Q1-95	1549.14	5	1	0	0	7.3454
7	Q2-95	2105.79	6	0	1	0	7.652
8	Q3-95	2041.32	7	0	0	1	7.6213
9	Q4-95	2021.01	8	0	0	0	7.6113
10	Q1-96	1870.46	9	1	0	0	7.5339
11	Q2-96	2390.56	10	0	1	0	7.7792
12	Q3-96	2198.03	11	0	0	1	7.6953
13	Q4-96	2046.83	12	0	0	0	7.6240
14	Q1-97	1934.19	13	1	0	0	7.5674
15	Q2-97	2406.41	14	0	1	0	7.7858

	A	B	C	D	E	F	G
7		Multiple		Adjusted	StErr of		
8	*Summary*	R	R-Square	R-Square	Estimate		
9		0.9628	0.9270	0.9218	0.102		
10							
11		Degrees of	Sum of	Mean of			
12	*ANOVA Table*	Freedom	Squares	Squares	F-Ratio	p-Value	
13	Explained	4	7.465	1.866	177.8172	< 0.0001	
14	Unexplained	56	0.588	0.010			
15							
16			Standard			Confidence Interval 95%	
17	*Regression Table*	Coefficient	Error	t-Value	p-Value	Lower	Upper
18	Constant	7.510	0.036	210.8236	< 0.0001	7.439	7.581
19	Time	0.019	0.001	25.9232	< 0.0001	0.018	0.021
20	Q1	-0.090	0.037	-2.4548	0.0172	-0.164	-0.017
21	Q2	0.140	0.037	3.7289	0.0005	0.065	0.215
22	Q3	0.091	0.037	2.4449	0.0177	0.017	0.166

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How to model this lagged features in sklearn?

- https://scikit-learn.org/stable/auto_examples/applications/plot_time_series_lagged_features.html

```
lagged_df = df.select(
    "count",
    *[pl.col("count").shift(i).alias(f"lagged_count_{i}h") for i in [1, 2, 3]],
    lagged_count_1d=pl.col("count").shift(24),
    lagged_count_1d_1h=pl.col("count").shift(24 + 1),
    lagged_count_7d=pl.col("count").shift(7 * 24),
    lagged_count_7d_1h=pl.col("count").shift(7 * 24 + 1),
    lagged_mean_24h=pl.col("count").shift(1).rolling_mean(24),
    lagged_max_24h=pl.col("count").shift(1).rolling_max(24),
    lagged_min_24h=pl.col("count").shift(1).rolling_min(24),
    lagged_mean_7d=pl.col("count").shift(1).rolling_mean(7 * 24),
    lagged_max_7d=pl.col("count").shift(1).rolling_max(7 * 24),
    lagged_min_7d=pl.col("count").shift(1).rolling_min(7 * 24),
)
lagged_df.tail(10)
```

month	sales	shift(1)	shift(1).rolling_mean(3)	shift(1).rolling_max(3)	shift(1).rolling_min(3)
1	100				
2	104	100			
3	112	104			
4	113	112	=AVERAGE(B2:B4)	112	100

(see model lagged term in sklearn.csv)

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Lecture 9

- ❑ Review
- ❑ **Autoregression**

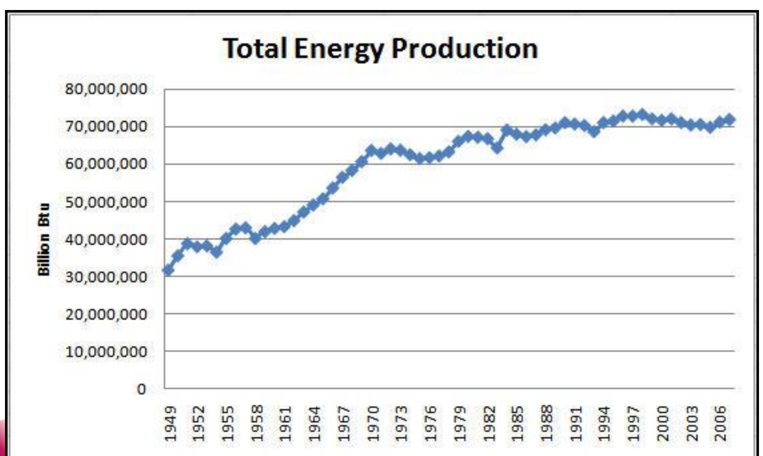
Reading Material:

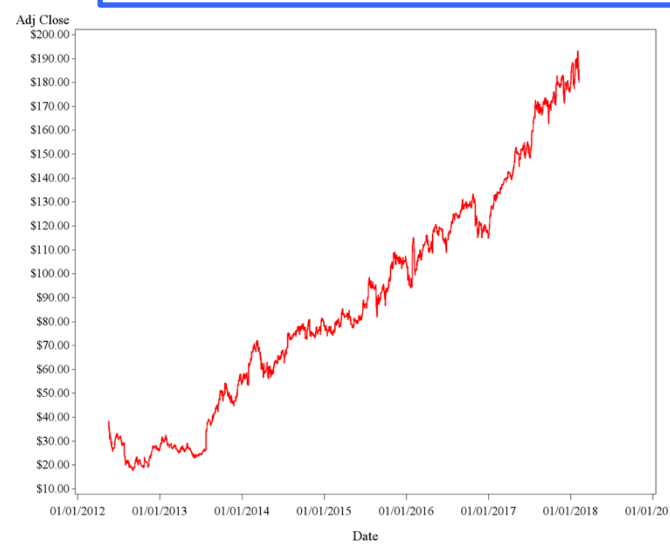
Introduction – Time Series

A ***Time Series*** is a set of observations of a variable at regular time intervals, such as yearly, monthly, weekly, daily, 15 minutes interval, etc.

- To predict the value of one dependent variable on the basis of time information
  - Time indexed Dependent Variable: **Yt**
  - Time indexed Independent Variable: **Xt**

How to get Xt?





Stock price of Facebook

Testing for Randomness

- **Typical Forecasting equation:**

$$Y_t = Fitted\ Value + Residual$$

- *We want this time series of residuals to be random noise.*
- *If the residual is not noise, it can be modeled further.* E.g. if the residual has an upward trend, we can modify the forecasting model to include this trend component in the *fitted* value.
- The point is that: the *fitted value* should include all components of the original series than can possibly be forecasted, and the leftover residuals should be unpredictable noise.

Autocorrelations

- Autocorrelation is the key to many forecasting methods, e.g. autoregressive models, especially more complex methods.
- It essentially specifies how observations in nearby time periods are related, so this information is often useful in forecasting.

For instance, the money in your bank account this month is highly related to your money last month.

Possible Xt?

Autocorrelations calculation

1. Forming lagged variables. Lags are simply previous observations, counting back to a certain number of periods from the present time.

For instance, Lag1 variables: $Y_t = Y_{t-1}$

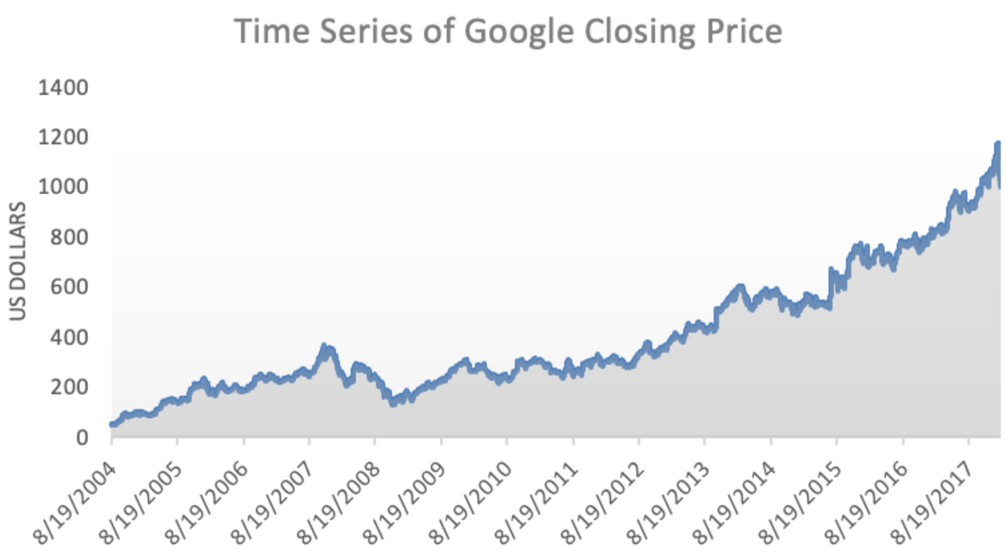
Lag 2 variables: $Y_t = Y_{t-2}$

2. Do the correlations between the observed values and the lagged variables.

	A	B	C	D	E	F	G	H	I	J
1	**Illustration of lagging and autocorrelation**									
2										
3	**Month**	**Sales**	**Sales_Lag1**	**Sales_Lag2**	**Sales_Lag3**		***Autocorrelations***			
4	Jan-95	226	.	.	.			Lag	Autocorr	StErr
5	Feb-95	254	226	.	.			1	**0.3492**	0.1443
6	Mar-95	204	254	226	.			2	0.0772	0.1443
7	Apr-95	193	204	254	226			3	0.0814	0.1443
8	May-95	191	193	204	254			4	-0.0095	0.1443
9	Jun-95	166	191	193	204			5	-0.1353	0.1443
10	Jul-95	175	166	191	193			6	0.0206	0.1443
11	Aug-95	217	175	166	191					
12	Sep-95	167	217	175	166					
48	Sep-98	175	181	179	168					
49	Oct-98	185	175	181	179					
50	Nov-98	245	185	175	181					
51	Dec-98	177	245	185	175					

The Random Walk Model

- The time series itself is not random (e.g. can see a trend in the visualized graph and autocorrelations are significantly positive for many lags).
- But *its differences* (the changes from one period to the next) **are random**.



Random Walk Model

$$Y_t = Y_{t-1} + m + e_t$$

Here:

Y_t - Observed value at the time period of t

Y_{t-1}- Observed value at the time period of t-1

M - the average differences

e_t - a random time series.

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Autoregressive Models

- We can easily extend the idea of random walk model by allowing coefficients in it.
- First-order autoregressive model AR(1)
  - $Y_t = a_0 + a_1 Y_{t-1} + e_t$
- We can further add more lag variables in the model.
- Second-order autoregressive model AR(2)
  - $Y_t = a_0 + a_1 Y_{t-1} + a_2 Y_{t-2} + e_t$
- …
- AR(p)
- You can use OLS to estimate model parameters

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ARIMA

Xt?

- AR = Auto Regressive
  - ✓ predict y_t from previous values $y_{t-1}, y_{t-2}, \ldots$

- MA = Moving Average
  - ✓ predict y_t from previous errors $\varepsilon_{t-1}, \varepsilon_{t-2}, \ldots$
    (Note: not the same as ETS MA.)

AR

- AR(1) model

$$y_t = c + \varphi y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim N(0, \sigma^2)$$

✓ $\varphi = 0$: white noise
✓ $\varphi = 1$ and $\mathrm{c} = 0$: random walk
✓ $\varphi = 1$ and $\mathrm{c} \neq 0$: random walk with a drift

AR

- AR(p) model

$$y_t = c + \varphi_1 y_{t-1} + \varphi_2 y_{t-2} + \cdots + \varphi_p y_{t-p} + \varepsilon_t$$

where $\varepsilon_t \sim N(0, \sigma^2)$

- How do we determine p?

Partial Autocorrelation Function (PACF)

In general, use x1,x2,x3 to predict y. The partial autocorrelation between y and x3:
1.Regression in which we predict *y* from x1 and x2,
2.regression in which we predict x3 from x1 and x2. Basically, we correlate the "parts" of *y* and x3 that are not predicted by x1 and x2.

For an AR model, the theoretical PACF "shuts off" past the order of the model.

AR

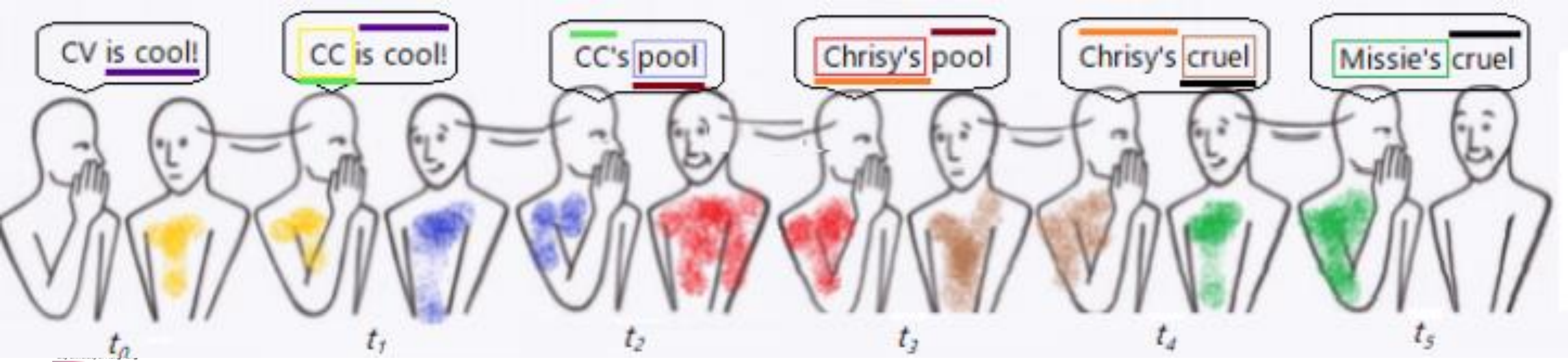
- AR(p) model

$$y_t = c + \varphi_1 y_{t-1} + \varphi_2 y_{t-2} + \cdots + \varphi_p y_{t-p} + \varepsilon_t$$

where $\varepsilon_t \sim N(0, \sigma^2)$

- How do we determine p?

Partial Autocorrelation Function (PACF)



MA

t	$\widehat{y_t}$	ε_t	y_t
1		-2	
2		1	
3		0	
4		2	
5		1	

MA(1) model

$$\boldsymbol{y_t = \mu + \phi_1 \varepsilon_{t-1} + \varepsilon_t},$$

where $\boldsymbol{\varepsilon_t \sim N(0, \sigma^2)}$

$$\boldsymbol{\widehat{y_t} = 10 + 0.5\varepsilon_{t-1}},$$

$$\boldsymbol{y_t = \widehat{y_t} + \varepsilon_t}$$

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MA

- MA(q) model

$$y_t = \mu + \phi_1 \varepsilon_{t-1} + \phi_2 \varepsilon_{t-2} + \cdots + \phi_q \varepsilon_{t-q} + \varepsilon_t$$

where $\varepsilon_t \sim N(0, \sigma^2)$

- How to determine q?

Autocorrelation Function (ACF)

ACF vs PACF

Predict average monthly price of salmon

P_{t-2}: avg. price 2 months ago

P_{t-1}: avg. price last month

P_t: avg. price this month

Correlation(P_{t-2}, P_t) = indirect effect + direct effect

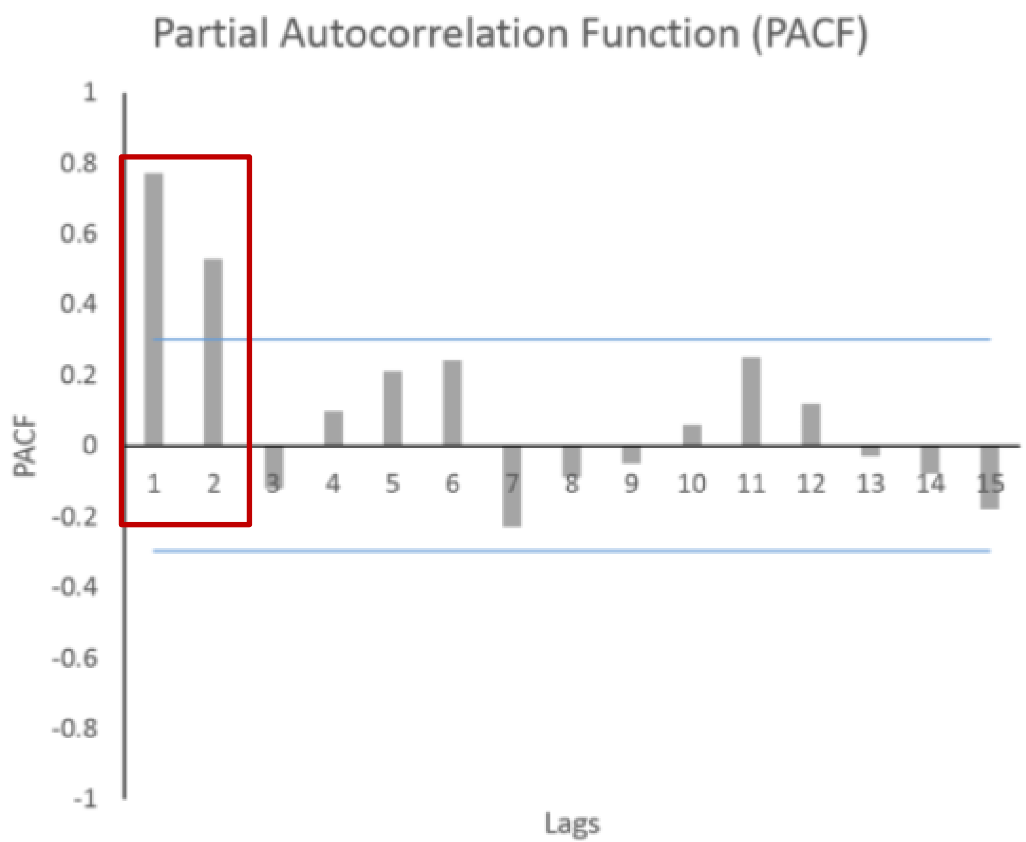
ACF **PACF**

AR & PACF

Two lags are significant

↓

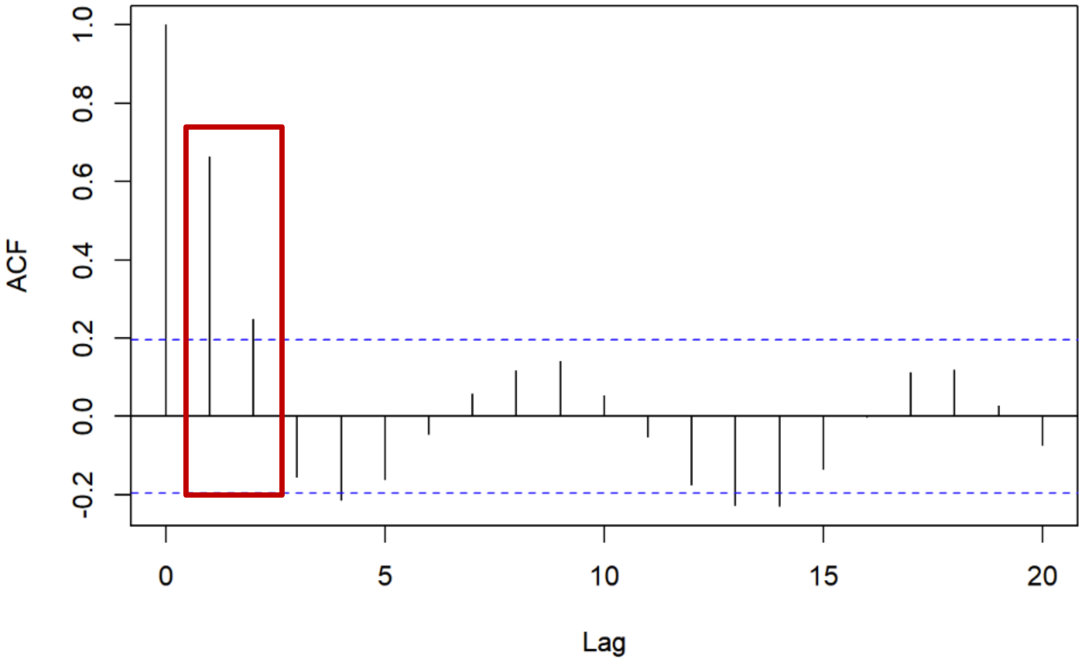
AR(2) model



MA & ACF

Two lags are significant

MA(2) model



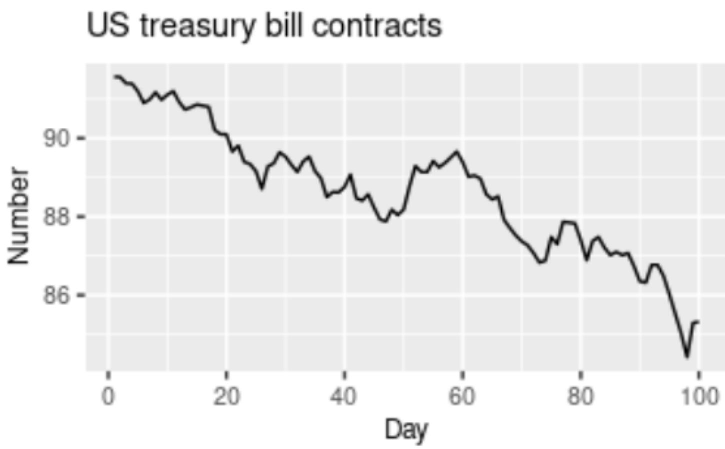
ARIMA

- AR = Auto-Regressive
  - ✓ predict y_t from previous values $y_{t-1}, y_{t-2}, \dots$
- MA = Moving Average
  - ✓ predict y_t from previous errors $\varepsilon_{t-1}, \varepsilon_{t-2}, \dots$
    (Note: not the same as ETS MA.)
- ARMA: assumes a stationary time series

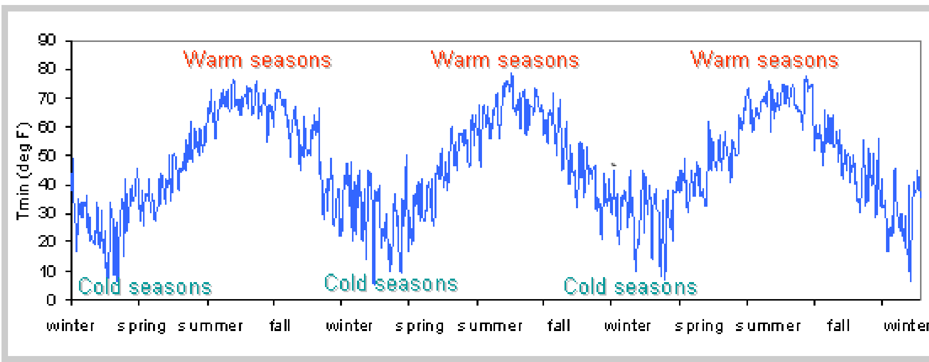
Stationarity

- A process is **stationary** if it has
  - ✓ No trends (i.e., constant mean)
  - ✓ Constant variance
  - ✓ No seasonality

Stationarity



A



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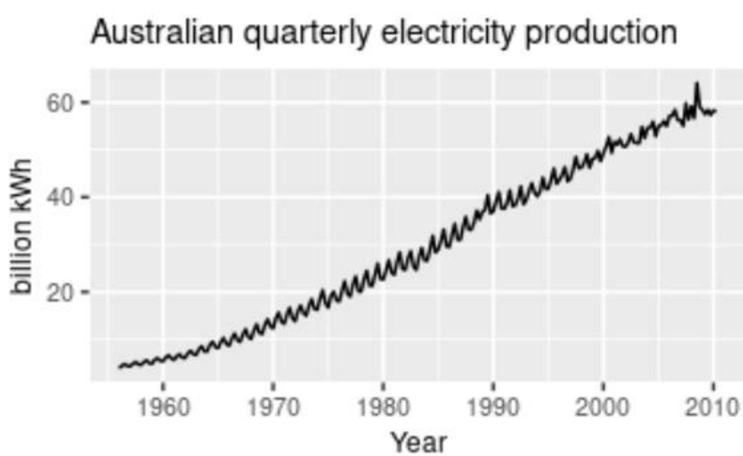
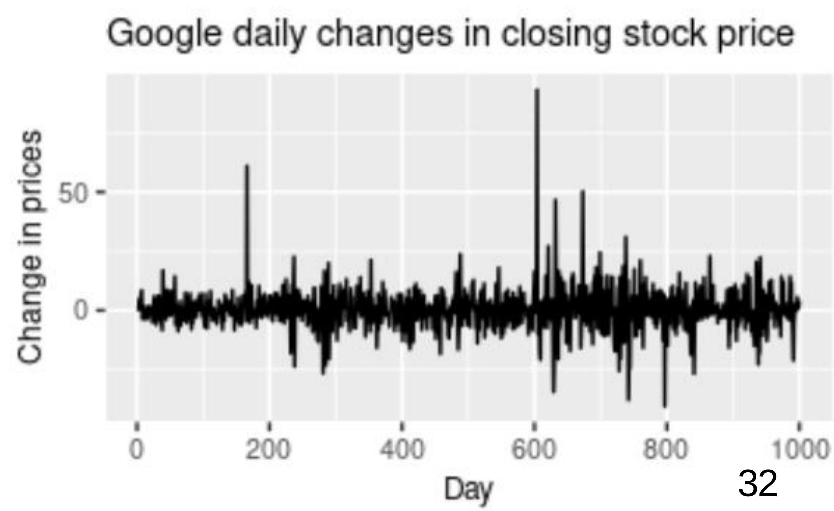


C



D

Stationarity test

- Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test

✓ Null hypothesis: data is stationary

✓ If p-value < 0.05, then reject the null, i.e., the data is non-stationary, at a confidence level of 95%. Otherwise, we accept that the data is stationary at a confidence level of 95%.

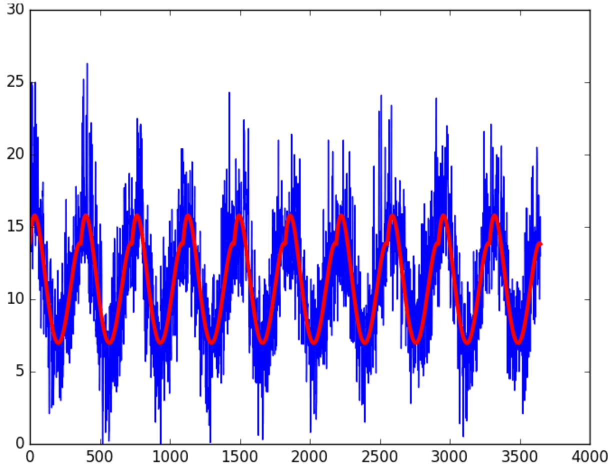
How to stabilize the data

- Stabilize variance
  - ✓ Box-Cox transformation

$$y(\lambda) = \begin{cases} \frac{y^\lambda - 1}{\lambda}, & \text{if } \lambda \neq 0; \\ \log y, & \text{if } \lambda = 0. \end{cases}$$

- De-trend
  - ✓ Differencing

- De-seasonality
  - ✓ Differencing
  - ✓ Subtract quarterly / monthly / etc. average



AR & Seasonal Effect

- You can apply AR after deseasonalization.
  - At the end, you should combine seasonal effect with AR model predictions for overall forecasting.

- If AR model contains enough lag variables, it may partially cover the effect of seasonal factors.
  - For example, if there is a yearly effect on quarterly data, including more than 4 lag variables (which covers things happened 4 quarters ago) may partially cover the surplus happens, say, every spring.

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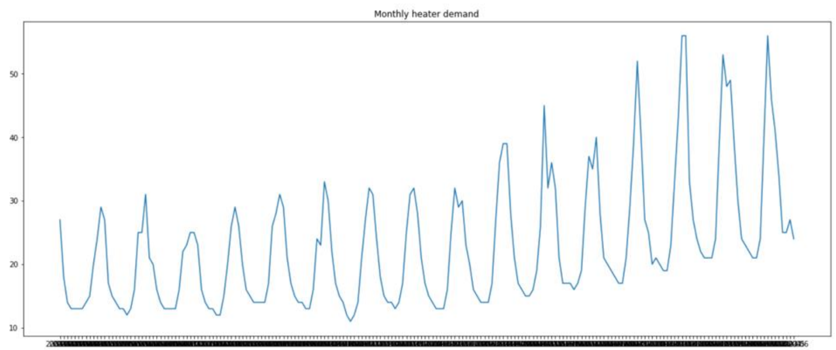
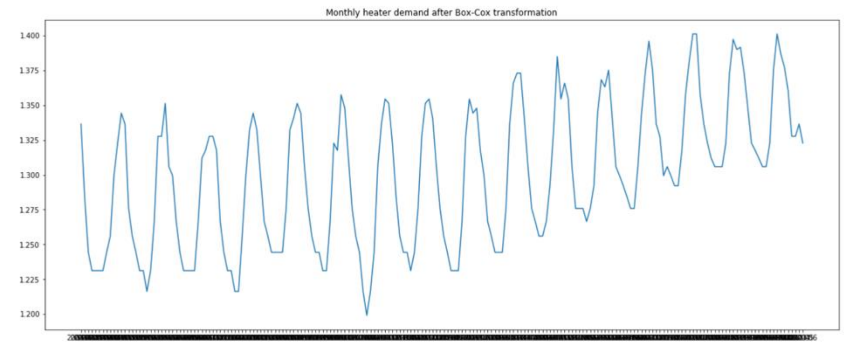
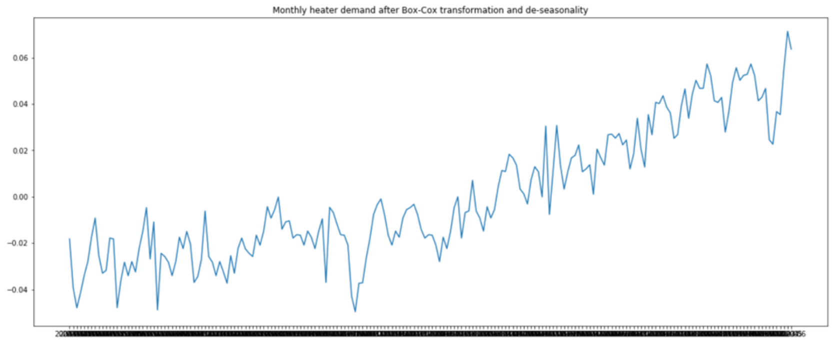
Quiz Question



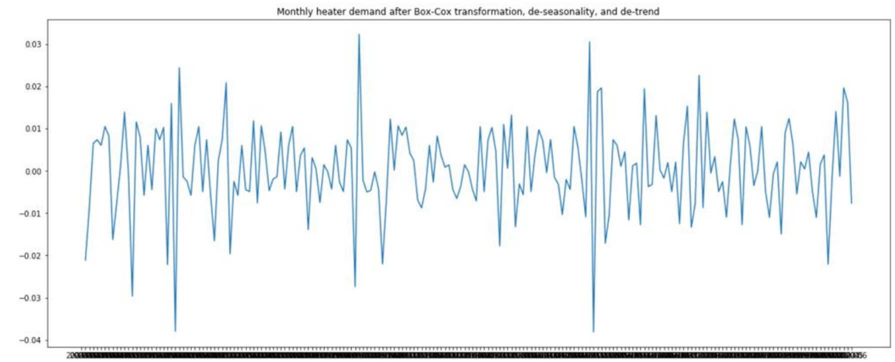
(a) Original data



(b) After Box-Cox Transform



(c) After BC transform and De-seasonality



(d) After BC transform, De-seasonality and De-trend.

Question:

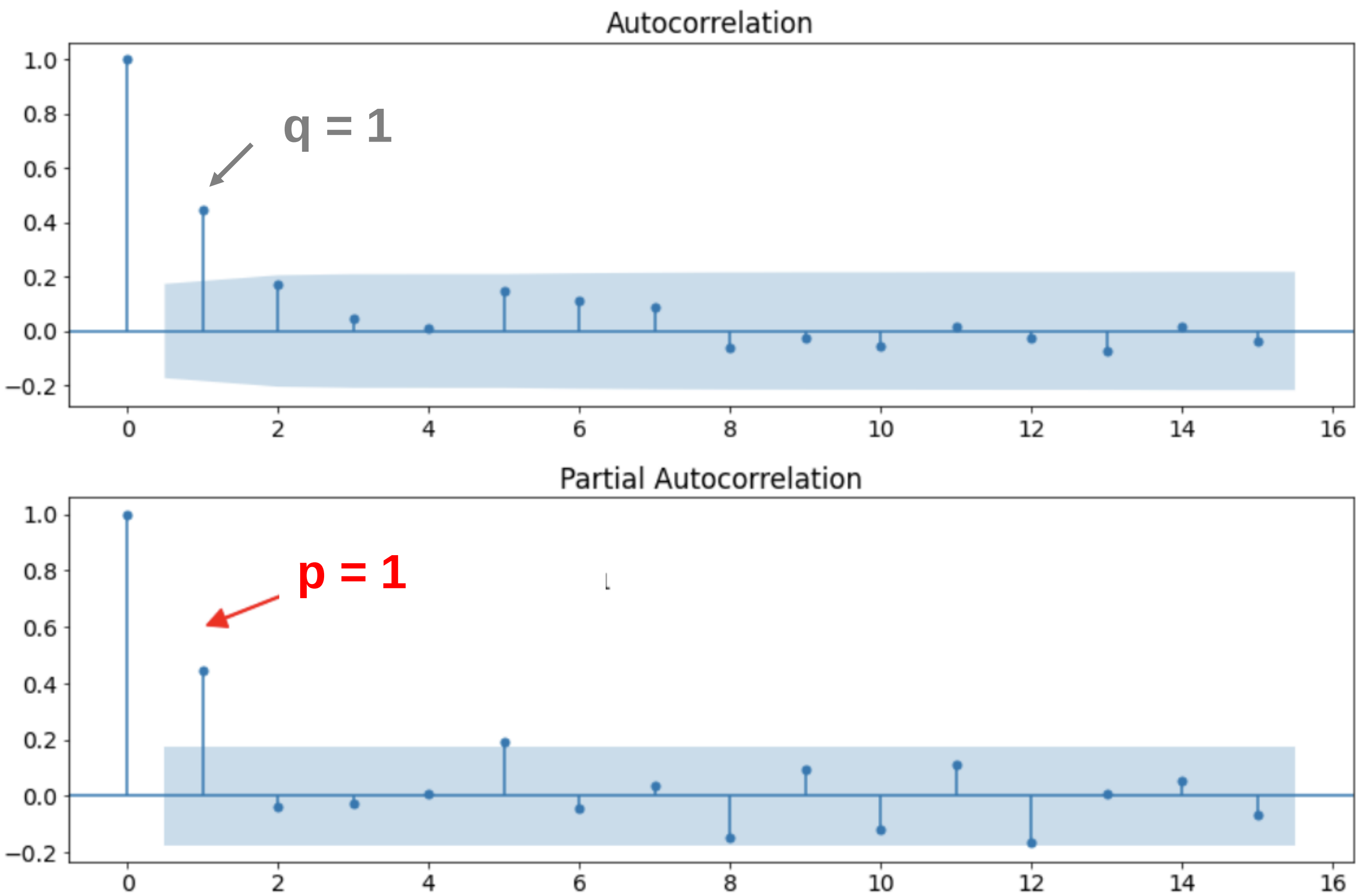
1. what did we do in (b), (c), and (d).
2. Is (a) or (d) stationary? How could you test stationarity?

ARMA

- ARMA(p, q)

$$y_t = c + \varphi_1 y_{t-1} + \varphi_2 y_{t-2} + \cdots + \varphi_p y_{t-p} + \phi_1 \varepsilon_{t-1} + \phi_2 \varepsilon_{t-2} + \cdots + \phi_q \varepsilon_{t-q} + \varepsilon_t$$

ARMA


Autocorrelation
q = 1
Partial Autocorrelation
p = 1
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0
2
4
6
8
10
12
14
16

ARMA

- ARMA(1, 1)

$$y_t = c + \varphi_1 y_{t-1} + \phi_1 \mathrm{\varepsilon}_{\mathrm{t-1}} + \mathrm{\varepsilon}_{\mathrm{t}}$$

ARIMA

- ARIMA(p, d, q): y is the d^{th}-differenced data

$$y'_t = c + \varphi_1 y'_{t-1} + \cdots + \varphi_p y'_{t-p} + \phi_1 \varepsilon_{t-1} + \cdots \phi_q \varepsilon_{t-q} + \varepsilon_t$$

✓ where y'_t is the d^{th}-differenced data
✓ when d = 1, $y'_t = y_t - y_{t-1}$
✓ when d = 2, $y''_t = y'_t - y'_{t-1} = y_t - 2y_{t-1} + y_{t-2}$

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ARIMA

- ARIMA(1,1,1)

$$y'_t = c + \varphi_1 y'_{t-1} + \phi_1 \varepsilon_{t-1} + \varepsilon_t$$

  ✓ where $y'_t = y_t - y_{t-1}$ and $y'_{t-1} = y_{t-1} - y_{t-2}$

- ARIMA(p,d,q)
  ✓ White noise: ARIMA(0,0,0)
  ✓ AR(p): ARIMA(p,0,0)
  ✓ MA(q): ARIMA(0,0,q)

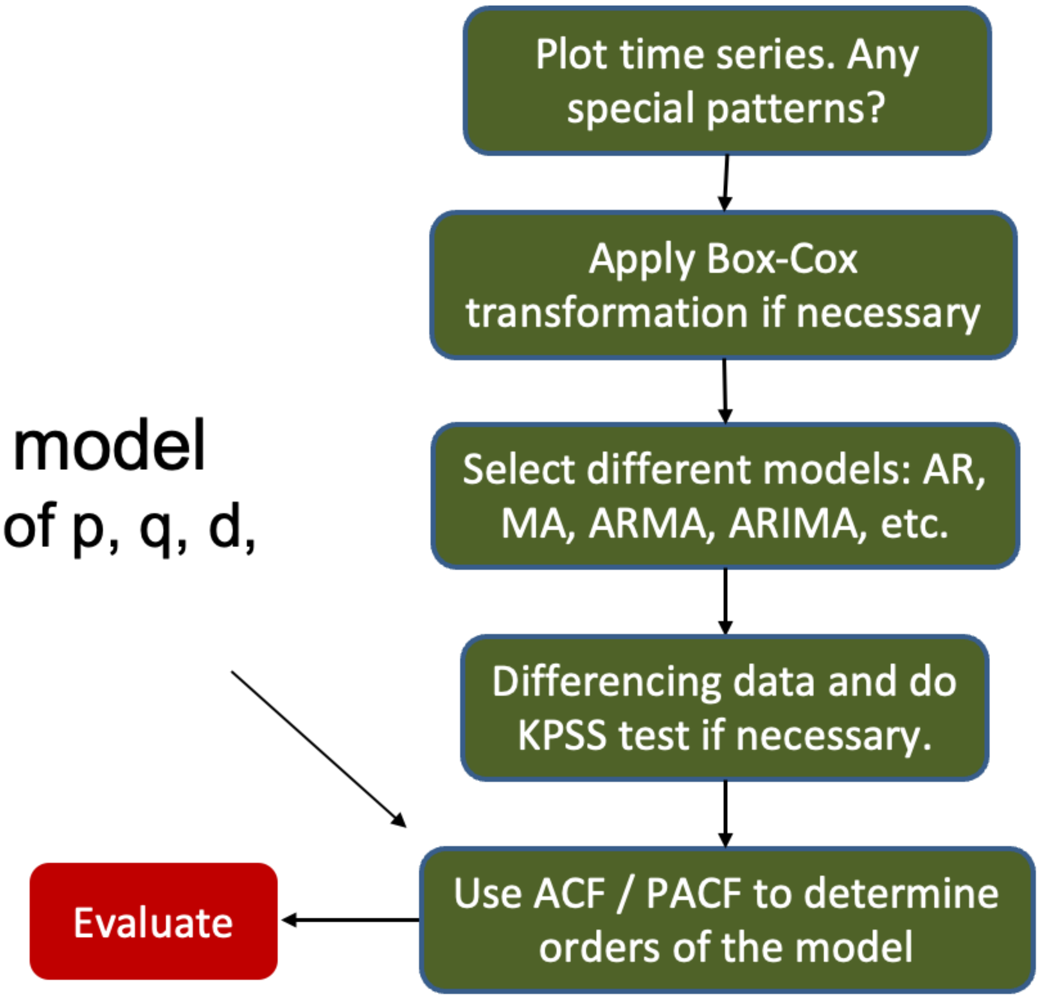
Seasonal ARIMA

$$\text{ARIMA} \quad \underbrace{(p, d, q)}_{\substack{\uparrow \\ \text{Non-seasonal part} \\ \text{of the model}}} \quad \underbrace{(P, D, Q)_m}_{\substack{\uparrow \\ \text{Seasonal part of} \\ \text{of the model}}}$$

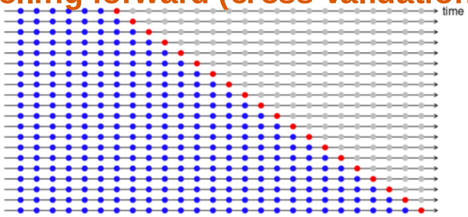
m: seasonal cycle (quarterly: 4, monthly: 12, weekly: 7, etc.)

Practical suggestions

- Prefer simpler model
  - ✓ Small values of p, q, d, P, Q, and D



Rolling forward (cross validation)

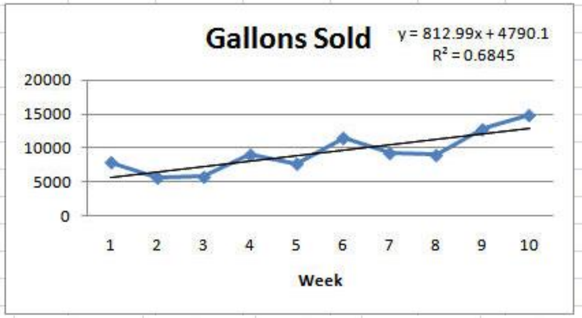


Regression With Causal Variables

- As in linear regression, we can incorporate causal factors to explain/predict things that may happen.

- A simple example
  - Sales = $\beta_0 + \beta_1$Week + β_2 Price/Gallon

Gasoline Sales		
Gallons Sold	Week	Price/Gallon
7815	1	$ 3.95
5541	2	$ 4.20
5650	3	$ 4.12
8949	4	$ 3.98
7600	5	$ 4.01
11430	6	$ 3.92
9190	7	$ 4.03
8889	8	$ 3.98
12721	9	$ 3.92
[illegible]	[illegible]	[illegible]



SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.930528528
R Square	0.865883342
Adjusted R Square	0.827564297
Standard Error	1235.400329
Observations	10

ANOVA

	df	SS	MS	F	Significance F
Regression	2	68974748.7	34487374.35	22.59668368	0.000883465
Residual	7	10683497.8	1526213.972		
Total	9	79658246.5			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	72333.08447	21969.92267	3.292368642	0.013259225	20382.47253	124283.6964
Week	508.6681395	168.1770861	3.024598364	0.019260863	110.9925233	906.3437558
Price/Gallon	-16463.19901	5351.082403	-3.076611005	0.017900405	-2911[illegible]	-3809.899789

ADL & Time Series Regression

- We can combine the idea of explanatory variables with AR model, i.e., make lag variables on explanatory variables.
- Autoregressive Distributed Lag
  - $Y_t = b_0 + b_1 Y_{t-1} + b_2 Y_{t-2} + \ldots + b_p Y_{t-p} +$ <= autoregressive
    $c_1 X_{t-1} + c_2 X_{t-2} + \ldots + c_q X_{t-q} + e_t$
  - ADL(p,q) distributed lag
- Time Series Regression w/ Multiple Predictors
  - $Y_t = b_0 + b_1 Y_{t-1} + b_2 Y_{t-2} + \ldots + b_p Y_{t-p} +$
    $c_1 X_{t-1} + c_2 X_{t-2} + \ldots + c_q X_{t-q} + \ldots$
    $d_1 Z_{t-1} + d_2 Z_{t-2} + \ldots + d_r Z_{t-r} + e_t$
- Parameter estimation
  - OLS
  - **Generalized OLS**

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Summary

- Stationarity
  - ✓ No trend, no seasonality, constant variance
- AR = Auto Regressive
  - ✓ Partial Autocorrelation Function (PACF)
- MA = Moving Average
  - ✓ Autocorrelation Function (ACF)
- ARMA / ARIMA / SARIMA
  - ✓ I(ntegrated): differencing